

four-axis positioning system item table

No.	Type	Item	Single-material Part / Assembly	Function	Composition	Materials	Need Rationale Source / Page / Verbatim Quote	Material Source / Page / Verbatim Quote
FS-1	Structure	Positioning base frame	Assembly	Support all motion-axis hardware and provide the structural datum relative to the chamber and electron beam.	Base frame; mounting bosses; datum surfaces; chamber mounting points; leveling pads.	Aluminum alloy; 304/316L stainless steel; titanium alloy; local Invar datum features if needed.	NASA EBF ³ patent says the positioning subsystem is structurally connected to the sealed container. Page p.9. Quote: “structurally connected to the sealed container.”	PI vacuum-positioning material guide lists typical vacuum positioning materials. Quote: “aluminum alloys, stainless steels, titanium, or bronze.”
FS-2	Structure	Axis mounting plate	Single-material plate or machined part	Provide a flat, stiff mounting surface for rails, screw supports, motors, and sensors.	Machined plate; tapped holes; dowel holes; cable pass-through holes.	Aluminum tooling plate; 304/316L stainless steel; titanium alloy for lightweight or stiff design.	NASA AIAA EBF ³ paper says the positioning system is attached to the chamber floor. Page p.3. Quote: “attached to the floor.”	PI vacuum-positioning guide supports aluminum, stainless steel, and titanium in vacuum positioning systems.
FS-3	Linear Motion	Linear guide rail	Single part	Provide straight, constrained travel for X, Y, and Z linear motion.	Precision rail; mounting holes; ground raceways.	Hardened stainless steel or bearing steel; stainless preferred for vacuum and corrosion robustness.	NASA AIAA EBF ³ paper states the positioning system has three orthogonal linear axes. Page p.3. Quote: “three orthogonal linear axes.”	Newport/MKS stage datasheet identifies “Linear ball bearings” in a precision stage.
FS-4	Linear Motion	Linear guide carriage	Assembly	Carry the moving axis plate along the linear guide rail.	Carriage body; recirculating balls; end caps; seals or shields.	Stainless steel carriage; ceramic or stainless balls; vacuum-compatible lubricant or dry film.	NASA patent says each axis includes translation hardware. Page p.9. Quote: “translation mechanism.”	NSK SPACEA catalog supports special-environment bearings and ball screws. Quote: “bearings, ball screws.”
FS-5	Linear Motion	Ball screw shaft	Single part	Convert motor rotation into precise linear displacement.	Ground or rolled screw shaft with helical raceway.	Hardened stainless steel or bearing steel; stainless option preferred for vacuum.	NASA patent explicitly lists ball screws as translation mechanisms. Page p.9. Quote: “ball screws.”	NSK SPACEA catalog supports ball screws for special environments. Quote: “bearings, ball screws.”
FS-6	Linear Motion	Ball nut	Assembly	Engage the ball screw and transmit linear force to the moving carriage.	Ball nut body; recirculating balls; return path; flange.	Stainless steel or bearing steel body; ceramic or stainless balls; vacuum-compatible lubricant.	NASA patent supports ball-screw translation mechanisms. Page p.9. Quote: “ball screws.”	Linear Motion Tips notes that recirculating bearing guides and screws can be made from stainless steel for vacuum applications.
FS-7	Linear Motion	Fixed-side screw bearing block	Assembly	Support one end of the ball screw and react axial thrust loads.	Bearing housing; angular-contact bearing pair;	Stainless steel housing; stainless or ceramic bearings; vacuum lubricant.	A ball screw axis requires screw support hardware; NASA patent	PI vacuum-positioning guide says special vacuum lubricants are

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					locknut; end cap.		supports ball-screw-based translation. Page p.9. Quote: “ball screws.”	used for high-vacuum and UHV motion systems. Quote: “special vacuum lubricants are used.”
FS-8	Linear Motion	Support-side screw bearing block	Assembly	Support the non-driven end of the ball screw and preserve screw alignment.	Bearing housing; radial bearing; retaining ring; end cap.	Stainless steel housing; ceramic or stainless bearing; vacuum lubricant.	NASA patent supports screw-based translation mechanisms. Page p.9. Quote: “translation mechanism.”	NSK SPACEA catalog supports special-environment bearings. Quote: “Hybrid Ceramic Linear Motion Ball Bearings.”
FS-9	Linear Motion	Axis motor	Assembly	Drive the ball screw or rotary drive under command from controls and power supplies.	Stepper or servo motor; rotor; stator; encoder if integrated; motor connector.	Vacuum-rated motor; stainless or aluminum housing; Kapton, PTFE, or PEEK wiring.	NASA patent says each axis includes a drive unit such as a stepper motor. Page p.9. Quote: “drive unit (Such as a stepper motor).”	PI vacuum-positioning guide lists vacuum cable insulation such as PTFE, FEP, Kapton, and PEEK.
FS-10	Linear Motion	Motor coupling	Assembly	Couple motor torque into the screw shaft while accommodating small misalignment.	Flexible coupling body; clamp hubs; set screws.	Stainless steel; aluminum alloy; bellows stainless coupling for vacuum precision use.	NASA patent requires drive units and translation mechanisms on each axis; a coupling is the mechanical interface between them. Page p.9. Quote: “drive unit.”	Material is engineering selection; final material should be locked by coupling vendor datasheet.
FS-11	Linear Motion	Motor mounting bracket	Single part or small assembly	Hold the axis motor in alignment with the screw drive.	Motor plate; standoffs; screws; alignment shoulder.	Aluminum alloy; 304/316L stainless steel; titanium if mass-critical.	NASA patent requires drive units on the axes. Page p.9. Quote: “drive unit.”	PI vacuum-positioning guide supports aluminum alloys, stainless steels, and titanium.
FS-12	Linear Motion	Linear encoder scale	Assembly	Provide direct position feedback for linear axes.	Scale; read head; mounting strip; signal cable.	Glass or metal scale; aluminum or stainless mount; shielded cable.	NASA patent lists rate and location sensors in the positioning subsystem. Page p.8. Quote: “rate and location sensors.”	Newport/MKS stage datasheet identifies screw-mounted rotary encoder and encoder feedback in stages.
FS-13	Linear Motion	Home sensor	Assembly	Establish a repeatable home reference for each axis.	Optical or magnetic sensor; target flag; bracket; cable.	Stainless or aluminum bracket; silicon sensor; Kapton or PTFE cable insulation.	NASA patent supports location sensing for the positioning subsystem. Page p.8. Quote: “location sensors.”	MKS/Newport vacuum stage datasheet lists “Origin Optical” on motorized stages.
FS-14	Linear Motion	Travel limit switch	Assembly	Prevent axis overtravel and support safety interlocks.	Mechanical or optical switch; target flag; bracket; cable.	Stainless or aluminum bracket; silicon or mechanical switch body;	NASA patent supports axis sensors and control of positioning. Page p.8.	MKS/Newport vacuum stage datasheet lists “Limit Switches” on

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						Kapton or PTFE cable.	Quote: “rate and location sensors.”	motorized stages.
FS-15	Z Axis	Z-axis brake	Assembly	Hold the vertical stage position when power is removed or during fault conditions.	Brake disk or electromagnetic brake; mount; release wiring.	Stainless steel brake hardware; copper coil if electromagnetic; friction material to be vacuum-verified.	NASA patent states the positioning subsystem has a Z linear axis and motorized positioning. Page p.9. Quote: “X, Y, Z linear axes.”	Material is brake-vendor specific; vacuum compatibility must be verified by supplier datasheet.
FS-16	Z Axis	Z-axis counterbalance spring	Single part or assembly	Reduce vertical load on the Z motor and screw drive.	Spring; anchor; adjustment screw; safety retainer.	Stainless spring steel; stainless fasteners.	This item is added for vertical-axis load management; NASA source establishes Z-axis motion but does not specify counterbalance. Page p.3. Quote: “150 mm in Z.”	Material is engineering selection; final spring material should be specified by spring vendor datasheet.
FS-17	Rotary Axis	Rotary table base	Single part or machined assembly	Support the rotary bearing and rotary drive on top of the linear axes.	Machined base; bearing seat; motor mount holes; cable passages.	Aluminum alloy; 304/316L stainless steel; titanium if mass-critical.	NASA AIAA EBF ³ paper states the fourth axis is rotary. Page p.3. Quote: “a rotary axis.”	PI vacuum-positioning guide supports aluminum, stainless steel, and titanium as vacuum mechanism materials.
FS-18	Rotary Axis	Cross-roller rotary bearing	Assembly	Provide stiff, precise rotational support for the build table.	Inner ring; outer ring; cross rollers; spacer cage; retaining hardware.	Stainless steel bearing rings; stainless or ceramic rollers; vacuum lubricant.	NASA AIAA EBF ³ paper states the rotary axis provides 360° rotation. Page p.3. Quote: “360° of rotation.”	Material is bearing-vendor specific; PI vacuum guide supports special vacuum lubricants. Quote: “special vacuum lubricants.”
FS-19	Rotary Axis	Worm wheel	Single part	Provide reduction and holding torque for the rotary axis.	Worm wheel gear only.	Aluminum bronze or stainless steel depending on load and galling risk.	NASA patent explicitly lists worm gears as a positioning mechanism. Page p.9. Quote: “worm gears.”	PI vacuum-positioning guide lists bronze and stainless steels as vacuum positioning materials.
FS-20	Rotary Axis	Worm shaft	Single part	Drive the worm wheel and rotate the build table.	Threaded worm shaft; bearing journals; motor-side end.	Hardened stainless steel or tool steel with vacuum-compatible surface finish.	NASA patent supports worm gears in the positioning system. Page p.9. Quote: “worm gears.”	Material is gear-vendor specific; final alloy should be locked by gear supplier datasheet.
FS-21	Rotary Axis	Rotary encoder	Assembly	Measure rotary-axis position for closed-loop control or verification.	Encoder disk or scale; read head; mount; signal cable.	Glass or metal encoder disk; aluminum or stainless mount; shielded cable.	NASA patent supports rate and location sensors for the positioning subsystem. Page p.8. Quote: “rate and location sensors.”	Newport/MKS stage datasheet lists rotary encoder feedback in motorized stages. Quote: “rotary encoder.”
FS-22	Build Interface	Moveable platform	Assembly	Carry the substrate or	Platform plate; mounting holes;	Aluminum tooling plate; stainless	NASA patent lists a moveable	PI vacuum-positioning

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				build plate through the programmed motion path.	locating features; electrical contact area.	steel; copper contact pad if needed.	platform in the positioning subsystem. Page p.8. Quote: “a moveable platform.”	guide supports aluminum alloys and stainless steels.
FS-23	Build Interface	Substrate clamp	Assembly	Secure the base plate or substrate during deposition.	Clamp jaw; screw; washer; locating pin.	Stainless steel clamp; Inconel or titanium for hot-zone use; ceramic washer if isolation is needed.	NASA patent lists clamping means for a base plate. Page p.8. Quote: “means for clamping a base plate.”	Material is engineering selection; final material should be selected by temperature, vacuum compatibility, and electrical isolation needs.
FS-24	Electrical / Grounding	Beam-current return strap	Assembly	Provide electrical continuity for beam-current return and measurement stability.	Flexible braid; lugs; fasteners; platform contact point.	Copper braid; tinned copper; stainless fasteners; copper contact pad.	NASA patent lists means for providing electrical continuity. Page p.8. Quote: “means for providing electrical continuity.”	Material is engineering selection; final grounding braid should be locked by supplier datasheet.
FS-25	Isolation	Thermal/electrical isolation block	Single part or small assembly	Thermally and electrically isolate positioning mechanisms from the process zone.	Insulating block; mounting screws; spacer washers.	Alumina ceramic; Macor; glass-ceramic; PEEK; G10/G11 depending on temperature and vacuum.	NASA patent states the positioning subsystem is thermally and electrically isolated from the process zone. Page p.9. Quote: “thermally and electrically isolated.”	PI vacuum-positioning guide supports PEEK in vacuum positioning systems. Quote: “PEEK.”
FS-26	Protection	Bellows cover	Assembly	Protect linear mechanisms from condensed metal vapor, debris, and process contamination.	Bellows; end plates; mounting screws; support brackets.	Stainless steel bellows or coated fabric only if vacuum-compatible; stainless preferred near hot/vacuum zones.	NASA patent states the mechanisms are covered with protective shielding. Page p.9. Quote: “protective shielding.”	Material is engineering selection; stainless steel is broadly supported for vacuum hardware by vacuum-material references.
FS-27	Protection	Sacrificial sheet-metal shield	Single part or small assembly	Protect motors, guides, screws, sensors, and cables from metal vapor and spatter.	Removable shield plate; standoffs; screws; overlap tabs.	304/316L stainless sheet; aluminum sheet; titanium sheet; Inconel near hot regions.	NASA patent says shielding avoids contamination with condensed metal vapor. Page p.9. Quote: “condensed metal vapor.”	PI vacuum-positioning guide supports stainless steels, aluminum alloys, and titanium.
FS-28	Cabling	Vacuum-compatible motor cable	Assembly	Route motor power inside the chamber without excessive outgassing.	Conductors; insulation; shield; strain relief; connector.	Copper conductors; PTFE/FEP, Kapton/polyimide, or PEEK insulation; braided shield.	NASA patent states electrical power is provided via feedthroughs to positioning axes. Page p.10. Quote: “electrical feed-through.”	PI vacuum-positioning guide lists PTFE/FEP, Kapton, and PEEK cable insulation for vacuum.
FS-29	Cabling	Vacuum-compatible	Assembly	Route encoder, home sensor,	Signal conductors;	Copper conductors;	NASA patent includes	PI vacuum-positioning

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		sensor cable		limit switch, and temperature signals inside the chamber.	insulation; shield; connector; strain relief.	Kapton/polyimide or PTFE/FEP insulation; braided shield.	sensors and electrical feed-throughs for motors/sensors. Page p.7. Quote: "sensors and motors."	guide supports vacuum cable insulation choices. Quote: "polyimide (Kapton) or PEEK."
FS-30	Chamber Interface	Electrical feedthrough interface	Assembly	Pass motor and sensor power/signals through the chamber wall while preserving vacuum.	Feedthrough flange; pins; ceramic insulator; external connector; internal connector.	Alumina ceramic; Kovar or stainless steel shell; copper or nickel pins.	NASA patent includes at least one electrical feed-through. Page p.7. Quote: "at least one electrical feed-through."	Lesker notes electrical feedthroughs cover many terminations, voltages, and current capacities.